

Product Requirements Document (PRD)

Simplified Network Monitoring System

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Status: Draft

Executive Summary

This document outlines the product requirements for a simplified network monitoring system inspired by PRTG Network Monitor. The system will provide real-time monitoring, alerting, and reporting capabilities for IT infrastructure including networks, servers, applications, and devices.

Product Vision

To create an intuitive, lightweight, and powerful network monitoring solution that enables businesses to:

- Monitor their IT infrastructure in real-time
 - Receive proactive alerts before issues become critical
 - Gain insights through comprehensive dashboards and reports
 - Scale from small businesses to medium enterprises
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Target Audience

Primary Users

- **IT Administrators:** Day-to-day monitoring and issue resolution
- **Network Engineers:** Network performance optimization
- **DevOps Engineers:** Application and service monitoring
- **System Administrators:** Server and infrastructure management

Secondary Users

- **IT Managers:** High-level reporting and analytics
 - **C-Level Executives:** Business impact visibility
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Core Features

1. Multi-Protocol Monitoring

1.1 Network Monitoring

- **Ping/ICMP Monitoring:** Device availability and response time
- **SNMP Monitoring:** Bandwidth, CPU, memory, interface statistics
- **NetFlow/sFlow:** Traffic analysis and bandwidth usage
- **Packet Sniffing:** Deep packet inspection for troubleshooting

1.2 Server Monitoring

- **CPU Usage:** Real-time and historical CPU utilization
- **Memory Usage:** RAM and swap space monitoring
- **Disk Space:** Storage capacity and I/O performance
- **Process Monitoring:** Critical service status tracking
- **System Uptime:** Availability tracking

1.3 Application Monitoring

- **HTTP/HTTPS Monitoring:** Website availability and response time
- **API Endpoint Monitoring:** RESTful API health checks
- **Database Monitoring:** MySQL, PostgreSQL, MongoDB query performance
- **Service Port Monitoring:** TCP/UDP port availability

1.4 Custom Monitoring

- **Script-Based Sensors:** Python, Bash, PowerShell custom scripts
 - **API Integration:** Third-party service monitoring via REST APIs
 - **Log File Monitoring:** Pattern matching and log analysis
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2. Real-Time Dashboards

2.1 Dashboard Components

- **Live Status Widgets:** Real-time metric displays
- **Charts and Graphs:** Line charts, bar charts, pie charts
- **Network Topology Maps:** Visual device relationship mapping
- **Gauge Displays:** CPU, memory, bandwidth gauges
- **Status Lists:** Sortable, filterable device lists

2.2 Dashboard Features

- **Customizable Layouts:** Drag-and-drop widget arrangement
 - **Multiple Dashboards:** Create task-specific views
 - **Auto-Refresh:** Configurable refresh intervals (5s - 5m)
 - **Full-Screen Mode:** NOC/SOC display optimization
 - **Dashboard Sharing:** Share via unique URLs
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3. Alerting System

3.1 Alert Triggers

- **Threshold-Based:** CPU > 80%, Memory > 90%, etc.
- **Status Change:** Device up/down state changes
- **Pattern-Based:** Log pattern detection
- **Absence-Based:** Missing heartbeat alerts
- **Composite Conditions:** Multiple condition combinations

3.2 Notification Channels

- **Email:** SMTP-based email notifications
- **SMS:** Twilio/SMS gateway integration
- **Webhooks:** HTTP POST to external systems
- **Slack/Teams:** Team collaboration tool integration
- **Mobile Push:** iOS/Android push notifications

3.3 Alert Management

- **Alert Escalation:** Multi-tier notification escalation
 - **Maintenance Windows:** Scheduled alert suppression
 - **Acknowledgement:** Manual alert acknowledgement
 - **Alert Dependencies:** Parent-child alert relationships
 - **Alert History:** Complete audit trail
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4. Reporting & Analytics

4.1 Report Types

- **Availability Reports:** Uptime/downtime percentages
- **Performance Reports:** Response time trends

- **Capacity Reports:** Resource utilization forecasting
- **SLA Reports:** Service level agreement compliance
- **Custom Reports:** User-defined report templates

4.2 Report Features

- **Scheduled Reports:** Daily, weekly, monthly automation
 - **Export Formats:** PDF, CSV, JSON, HTML
 - **Historical Data:** 30/90/365 day comparisons
 - **Trend Analysis:** Performance trend visualization
 - **Report Templates:** Pre-built report designs
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5. User Management & Security

5.1 Authentication

- **Local Authentication:** Username/password
- **LDAP/Active Directory:** Enterprise SSO integration
- **Two-Factor Authentication:** TOTP-based 2FA
- **API Keys:** REST API authentication tokens

5.2 Authorization

- **Role-Based Access Control (RBAC):** Admin, Operator, Viewer
 - **Resource-Level Permissions:** Device/group access control
 - **Custom Roles:** User-defined role creation
 - **Audit Logging:** User action tracking
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6. Auto-Discovery

6.1 Discovery Methods

- **Network Scanning:** IP range scanning (ICMP, SNMP)
- **Active Directory Discovery:** Windows domain integration
- **Cloud Provider APIs:** AWS, Azure, GCP auto-discovery
- **VMware vCenter:** Virtual infrastructure discovery

6.2 Discovery Configuration

- **Scheduled Discovery:** Automatic periodic scanning
 - **Discovery Templates:** Pre-configured scan profiles
 - **Device Classification:** Automatic device categorization
 - **Template Assignment:** Auto-sensor creation
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Technical Requirements

Performance Requirements

- **Scalability:** Support 1000+ monitored devices
- **Data Collection:** 10-second minimum polling interval
- **Dashboard Load Time:** < 2 seconds
- **Alert Latency:** < 30 seconds from trigger to notification
- **Data Retention:** 1 year historical data

Reliability Requirements

- **System Uptime:** 99.9% availability
- **Data Accuracy:** 99.99% metric accuracy
- **Failover Support:** High availability clustering
- **Backup & Recovery:** Automated configuration backups

Compatibility Requirements

- **Operating Systems:** Windows, Linux, macOS
 - **Browsers:** Chrome, Firefox, Safari, Edge (latest 2 versions)
 - **Mobile Platforms:** iOS 14+, Android 10+
 - **Database:** PostgreSQL 12+, MySQL 8+
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User Interface Requirements

Design Principles

- **Responsive Design:** Mobile-first approach
- **Dark Mode:** Optional dark theme
- **Accessibility:** WCAG 2.1 AA compliance
- **Modern Aesthetics:** Clean, professional interface
- **Intuitive Navigation:** Minimal learning curve

Key Screens

1. **Login Screen:** Secure authentication portal
 2. **Main Dashboard:** Overview of all monitored systems
 3. **Device Detail View:** Comprehensive device metrics
 4. **Alert Management:** Alert list and configuration
 5. **Report Builder:** Interactive report creation
 6. **Settings Panel:** System configuration interface
 7. **User Management:** User and role administration
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Data Requirements

Metrics Storage

- **Time-Series Database:** Efficient metric storage
- **Compression:** Data compression for storage optimization
- **Aggregation:** Automatic data rollup (hourly, daily)
- **Purging:** Configurable data retention policies

Configuration Storage

- **Device Database:** Device inventory and settings
 - **User Database:** User accounts and permissions
 - **Alert Rules:** Alert configuration persistence
 - **Templates:** Monitoring templates library
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Integration Requirements

APIs

- **RESTful API:** Complete CRUD operations
- **GraphQL API:** Flexible query interface
- **WebSocket API:** Real-time data streaming
- **Webhook Support:** Outbound event notifications

Third-Party Integrations

- **Ticketing Systems:** Jira, ServiceNow integration
- **Collaboration Tools:** Slack, Microsoft Teams
- **Cloud Platforms:** AWS CloudWatch, Azure Monitor
- **Configuration Management:** Ansible, Puppet, Chef

Non-Functional Requirements

Usability

- **Setup Time:** < 30 minutes for basic setup
- **Learning Curve:** Productive within 2 hours
- **Documentation:** Comprehensive user guides
- **Help System:** Context-sensitive help

Maintainability

- **Monitoring Agent Updates:** Automatic update mechanism
- **Database Migrations:** Zero-downtime schema updates
- **Plugin Architecture:** Extensible sensor framework
- **Configuration Export/Import:** Easy migration

Security

- **Encryption:** HTTPS/TLS 1.3 for all communications
- **Data Protection:** Database encryption at rest
- **Input Validation:** SQL injection prevention
- **Session Management:** Secure session handling
- **Vulnerability Scanning:** Regular security audits

Success Metrics

Key Performance Indicators (KPIs)

- **Mean Time to Detection (MTTD):** < 1 minute
- **Mean Time to Resolution (MTTR):** 20% reduction
- **False Positive Rate:** < 5%
- **User Adoption Rate:** > 80% within 3 months
- **System Uptime:** > 99.9%
- **Customer Satisfaction:** > 4.5/5 stars

Release Strategy

Phase 1: MVP (Months 1-3)

- Basic ping/SNMP monitoring
- Simple dashboard
- Email alerting
- User authentication

Phase 2: Enhanced (Months 4-6)

- Advanced monitoring protocols
- Custom dashboards
- Multiple notification channels
- Basic reporting

Phase 3: Enterprise (Months 7-9)

- Auto-discovery
- Advanced analytics
- RBAC and SSO
- API integration

Phase 4: Advanced (Months 10-12)

- AI-powered anomaly detection
 - Predictive analytics
 - Mobile apps
 - Advanced integrations
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Out of Scope

The following features are explicitly out of scope for the initial release:

- Network configuration management
 - Automated remediation actions
 - NetFlow collector (deferred to Phase 2)
 - Mobile application (deferred to Phase 4)
 - AI/ML-based predictions (deferred to Phase 4)
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Assumptions & Dependencies

Assumptions

- Users have basic networking knowledge
- Monitored devices support standard protocols (SNMP, HTTP)
- Network infrastructure allows monitoring traffic
- Sufficient server resources for time-series data storage

Dependencies

- PostgreSQL or MySQL database availability
 - SMTP server for email notifications
 - Network access to monitored devices
 - Modern web browser availability
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Risks & Mitigation

Risk	Impact	Probability	Mitigation
Complex SNMP configurations	High	Medium	Provide templates and wizards
High data volume	High	High	Implement data aggregation and compression
False positive alerts	Medium	High	Intelligent thresholding and baselines
Scalability bottlenecks	High	Medium	Distributed architecture design
Security vulnerabilities	High	Low	Regular security audits and updates

Glossary

- **Sensor:** A monitoring unit that collects specific metrics
- **Device:** Any monitored IT asset (server, router, switch, etc.)
- **Probe:** Data collection agent
- **Channel:** Individual data point within a sensor
- **Alert:** Notification triggered by threshold violation
- **Dashboard:** Visual representation of monitoring data
- **Template:** Pre-configured monitoring profile

- **Node:** Generic term for monitored entity
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Appendix

A. Competitor Analysis

- **PRTG Network Monitor:** Feature-rich but expensive
- **Zabbix:** Open-source but complex setup
- **Nagios:** Powerful but dated interface
- **Datadog:** Modern but cloud-only

B. Technology Stack Recommendations

- **Frontend:** React, TypeScript, TailwindCSS
- **Backend:** Node.js, Python (FastAPI)
- **Database:** PostgreSQL + TimescaleDB
- **Real-time:** WebSocket, Redis
- **Monitoring Agents:** Go, Python